

THE SHIFTED ZETA STRING: AN UNCONDITIONAL INVERSE-SPECTRAL FAMILY AND ITS RH ENDPOINT

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ABSTRACT. Suzuki’s shifted screw functions Ψ_ω attach to the Riemann zeta-function, for every $\omega \geq \frac{1}{2}$ *unconditionally* — and for every $\omega > \Theta - \frac{1}{2}$ under the corresponding zero-free half-plane — a positive inverse-spectral system: a Krein string $S[m_\omega, \infty)$ whose principal Weyl function is $q_\omega(z) = i(\xi'/\xi)(\frac{1}{2} + \omega - i\sqrt{z})/\sqrt{z}$. We identify its spectral measure in closed form: it is purely absolutely continuous, and its density is the zero-counting measure of ζ smeared by Poisson kernels, each zero $\rho = \beta + i\tau$ contributing a Lorentzian of width $\frac{1}{2} + \omega - \beta$ centred at τ — of width exactly ω under the Riemann Hypothesis. Three exact bridges follow. The screw function itself is the spectral transform $\Psi_\omega(t) = \int_{\mathbb{R}} (1 - \cos ut) u^{-2} \rho_\omega(u) du$; consequently the linear margin of the shifted screw criterion, studied in the companion paper, equals π times the spectral density at zero frequency. The string has a homogeneous infrared bulk whose density is the inverse square of that margin, $\rho_\infty(\omega) = [(\xi'/\xi)(\frac{1}{2} + \omega)]^{-2}$, and a universal ultraviolet law $m_\omega(x) \sim 4x(\log x)^{-2}$, the unconditional analogue of Kotani’s theorem for his RH-conditional zeta string. Finally, the family detects the zeros through its existence: q_ω is a Stieltjes function for every $\omega > 0$ if and only if the Riemann Hypothesis holds — a zero off the critical line creates non-real poles, destroying the positive string realisation below the critical shift — and under RH the family degenerates at its endpoint onto Kotani’s string: the spectral measures converge vaguely, as $\omega \downarrow 0$, to the pure-point measure $2 \sum_{\tau > 0} \delta_{\tau^2}$, with locally uniform convergence of Weyl functions at rate $O(\omega)$. The degeneration is geometric as well as spectral: every ω -string has infinite length, while Kotani’s limit string has finite length $(\xi'/\xi)'(\frac{1}{2}) = 0.04620\dots$. We provide independent numerical checks of the spectral formulas and compute finite approximations to the associated strings by discretisation and Herglotz peeling.

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1. INTRODUCTION

1.1. Context. Suzuki [15] attached to the Riemann zeta-function a continuous real function $\Psi(t)$ — the twice-integrated, normalised error term in the prime number theorem — and proved that the Riemann Hypothesis (RH) is equivalent to $\Psi \geq 0$ pointwise. The mechanism is the Kreĭn–Langer theory of screw functions [10]: RH holds if and only if $Q_\xi(z) = i(\xi'/\xi)(\frac{1}{2}-iz)$ is a Herglotz function (Lagarias [11]), if and only if $-\Psi$ is a screw function. Under RH, the associated Stieltjes-class function $q_\xi(z) = Q_\xi(\sqrt{z})/\sqrt{z}$ has the expansion $\sum_{\tau>0} 2/(\tau^2-z)$ over the ordinates of the zeros, and Kreĭn’s inverse theorem [9] attaches to it a string — Kotani’s *zeta string* [7], with pure point spectral measure of mass 2 at each τ^2 and mass function $m_\xi(x) \sim 4x(\log x)^{-2}$ as $x \rightarrow 0^+$ ([7]; see [15, §9]).

All of that is conditional: without RH there is no positive object at $\omega = 0$. Suzuki also introduced [15, §11] the shifted family Ψ_ω , with

$$\int_0^\infty \Psi_\omega(t) e^{izt} dt = -\frac{1}{z^2} \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega - iz \right), \quad (1)$$

and proved that $\xi(s) \neq 0$ on $\text{Re}(s) > \frac{1}{2} + \omega$ if and only if $\Psi_\omega \geq 0$ eventually. The companion paper [13] makes that criterion quantitative: unconditionally

$$\Psi_\omega(t) = \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega \right) t + \left(\frac{\xi'}{\xi} \right)' \left(\frac{1}{2} + \omega \right) + O\left(e^{-(\omega-\vartheta)t} \right), \quad \vartheta = \Theta - \frac{1}{2}, \quad (2)$$

with explicit constants; the linear coefficient — the *margin slope* — vanishes as $\omega \rightarrow 0^+$ precisely because $(\xi'/\xi)(\frac{1}{2}) = 0$, which is why RH is the marginal member of the family. For $\omega \geq \frac{1}{2}$ the criterion is unconditional — $\zeta(s) \neq 0$ on $\text{Re}(s) > 1$ is the prime number theorem — and, as recalled in [13], Suzuki’s Herglotz argument then makes $-\Psi_\omega$ a screw function outright. Combining that shifted Herglotz positivity with Kreĭn’s correspondence suggests that ζ carries, unconditionally, a one-parameter family of positive inverse-spectral systems, extending Suzuki’s construction at $\omega = 0$ [15, §9]. The present paper develops that family explicitly.

1.2. Results. Fix $\omega \geq \frac{1}{2}$ and write $\sigma = \frac{1}{2} + \omega$. For $z \in \mathbb{C} \setminus [0, \infty)$ take the branch of $\sqrt{z}$ with $\text{Im} \sqrt{z} > 0$ and set $w = -i\sqrt{z}$, so $\text{Re} w > 0$ and $z = -w^2$. Define

$$q_\omega(z) := \frac{(\xi'/\xi)(\sigma + w)}{w} = \frac{i}{\sqrt{z}} \left(\frac{\xi'}{\xi} \right) (\sigma - i\sqrt{z}). \quad (3)$$

Zeros of ξ are written $\rho = \beta + i\tau$; sums over ρ count multiplicity. Recall the Stieltjes class $\mathcal{N}_S$: Herglotz functions analytic and non-negative on $(-\infty, 0)$, i.e. functions of the form $b + \int_0^\infty (\lambda - z)^{-1} d\sigma(\lambda)$ with $b \geq 0$ and $\int (1 + \lambda)^{-1} d\sigma < \infty$; Kreĭn’s correspondence [9, 3] attaches to each such function a unique string $S[m, L]$ (a length $0 < L \leq \infty$ and a non-decreasing mass function m on $[0, L]$) whose principal Titchmarsh–Weyl function it is.

Theorem 1.1 (spectral measure; Section 3). *Let $\omega \geq \frac{1}{2}$. Then $q_\omega \in \mathcal{N}_S$, with $b = 0$, no linear term, and purely absolutely continuous spectral measure*

$$d\sigma_\omega(\lambda) = \frac{\rho_\omega(\sqrt{\lambda})}{\sqrt{\lambda}} d\lambda \quad \text{on } [0, \infty), \quad \rho_\omega(u) = \frac{1}{\pi} \text{Re} \left(\frac{\xi'}{\xi} \right) (\sigma + iu) = \frac{1}{\pi} \sum_\rho \frac{\sigma - \beta}{(\sigma - \beta)^2 + (u - \tau)^2}, \quad (4)$$

the sum over all zeros being absolutely convergent. Consequently there is a unique string $S[m_\omega, L_\omega]$ with principal Weyl function q_ω ; it has $L_\omega = \infty$ and $m_\omega(\infty) = \infty$. The same statements hold for $\omega \in (\vartheta, \frac{1}{2})$ conditionally on the zero-free half-plane $\text{Re}(\rho) \leq \frac{1}{2} + \vartheta < \sigma$.

The density (4) is the zero-counting measure smeared by Poisson kernels: each zero contributes a Lorentzian of width $\sigma - \beta$ — its distance to the boundary line — centred at its ordinate. Under RH every width equals ω exactly. See Figure 1.

Theorem 1.2 (screw representation and the margin identity; Section 4). *Let $\omega \geq \frac{1}{2}$ (or $\omega > \vartheta$ conditionally). Then for all $t \geq 0$,*

$$\Psi_\omega(t) = \int_{\mathbb{R}} \frac{1 - \cos(ut)}{u^2} \rho_\omega(u) du, \quad (5)$$

absolutely convergent, with exactly the stated normalisation. In particular

$$\boxed{\left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right) = \pi \rho_\omega(0)} \quad (6)$$

— *the margin slope of [13] is π times the spectral density at zero frequency.*

Theorem 1.3 (infrared bulk; Section 5). *Under the hypotheses of Theorem 1.1 (in particular, unconditionally for $\omega \geq \frac{1}{2}$),*

$$\frac{m_\omega(x)}{x} \longrightarrow \rho_\infty(\omega) := \left[\left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right)\right]^{-2} \quad (x \rightarrow \infty).$$

At $\omega = \frac{1}{2}$ the bulk density of the unconditional zeta string is $(\xi'/\xi)(1)^{-2} = 1874.7243\dots$, with $(\xi'/\xi)(1) = \frac{\gamma_0}{2} + 1 - \frac{1}{2} \log 4\pi = 0.0230957\dots$

Together, (6) and Theorem 1.3 give the chain that motivates this paper:

$$\text{margin slope} = \pi \times (\text{zero-frequency density of states}) = \rho_\infty^{-1/2}.$$

The central quantity of the criterion in [13] is, on the string side, the reciprocal square root of the bulk density.

Theorem 1.4 (ultraviolet law; Section 5). *Under the hypotheses of Theorem 1.1,*

$$m_\omega(x) \sim \frac{4x}{(\log x)^2} \quad (x \rightarrow 0^+);$$

in particular this holds unconditionally for every $\omega \geq \frac{1}{2}$, with the same law and constant as Kotani's RH-conditional zeta string [7]. (The asymptotic computation on the Weyl-function side is unconditional and ω -independent throughout; only the existence of the positive string requires the hypothesis.)

The last group of results concerns the endpoint $\omega \downarrow 0$, and is where the family sees RH.

Theorem 1.5 (existence of the family $\iff$ RH; Section 6). *Let $0 < \omega$ and suppose ξ has a zero $\rho = \beta + i\tau$, $\tau > 0$, with $\beta > \sigma = \frac{1}{2} + \omega$. Then q_ω is meromorphic on $\mathbb{C} \setminus [0, \infty)$ with the non-real pole pair*

$$z_\rho = \tau^2 - (\beta - \sigma)^2 \mp 2i\tau(\beta - \sigma) \quad (\text{from } \rho \text{ and } \bar{\rho}),$$

hence $q_\omega \notin \mathcal{N}_S$ and no positive string realisation exists. Combined with Theorem 1.1:

$$\boxed{RH \iff q_\omega \in \mathcal{N}_S \text{ for every } \omega > 0 \iff \text{the } \omega\text{-string family extends to all } \omega > 0.}$$

Theorem 1.6 (RH endpoint; Section 6). *As $\omega \downarrow 0$:*

- (i) *Unconditionally, $q_\omega \rightarrow q_\xi$, $q_\xi(z) = i(\xi'/\xi)(\frac{1}{2} - i\sqrt{z})/\sqrt{z}$, locally uniformly on $\mathbb{C} \setminus ([0, \infty) \cup P)$, where P is the pole set of q_ξ ; the rate is $O(\omega)$, locally uniformly. Under RH, $P \subset [0, \infty)$ and the convergence is locally uniform on all of $\mathbb{C} \setminus [0, \infty)$.*

(ii) Under RH, the spectral measures converge vaguely to Kotani's pure-point measure:

$$\sigma_\omega \implies 2 \sum_{\tau > 0} \delta_{\tau^2} \quad (\omega \downarrow 0).$$

(iii) Under RH, $m_\omega(x) \rightarrow m_\xi(x)$ at every continuity point of m_ξ , by continuity of Kreĭn's correspondence [4, Thm. 1].

Without RH, by Theorem 1.5 the family cannot be continued below $\omega_c = \Theta - \frac{1}{2} > 0$ (for $\omega = \omega_c$ itself see Remark 6.1) and the endpoint is never reached: the statement “ σ_ω converges, as $\omega \downarrow 0$, to a positive measure” is itself equivalent to RH at the level of the family.

Under RH the transition at $\omega = 0$ is thus an absolutely-continuous-to-pure-point spectral limit: the Lorentzians of Theorem 1.1, all of width exactly ω , sharpen to the point masses of Kotani's string. The endpoint is singular geometrically as well: every ω -string has infinite length, while Kotani's limit string has finite length $(\xi'/\xi)'(\frac{1}{2})$ (Remark 6.3). We do not claim any statement about eigenfunctions or dynamics, and accordingly avoid the word “localisation” except as an interpretive gloss (Section 8).

1.3. What is new, and what is Suzuki's. The objects are Suzuki's: the screw function, the shifted family, the Laplace identity (1), and, under RH, the identification of q_ξ with Kotani's string [15, §9]; the margin asymptotics (2) are from the companion paper [13]. What is offered here is the unconditional inverse-spectral family itself: the closed-form identification of the spectral measure (Theorem 1.1), the exact normalisation of the spectral representation of Ψ_ω and the resulting margin identity (Theorem 1.2), the bulk and ultraviolet laws for the string (Theorems 1.3, 1.4), the equivalence of RH with the existence of the full family and the endpoint degeneration onto Kotani's string (Theorems 1.5, 1.6), and the numerical computation of the strings (Section 7).

Source of the Tauberian input. Theorems 1.3 and 1.4 reduce, after exact computations with q_ω , to Kasahara's regular-variation theorem for Kreĭn's correspondence, [4, Thm. 2 and Example 1], applied at index $\alpha = \frac{1}{2}$, where his constant is $D_{1/2} = 1$; Theorem 1.6(iii) uses his continuity theorem [4, Thm. 1]. We have checked the statements, constants, and endpoint conventions against the original (see also [8, 5, 6] and, for canonical systems, [2]). Two convention notes: Kasahara's characteristic function $h(s) = a + \int_0^\infty (\lambda + s)^{-1} d\sigma(\lambda)$ [4, (5), (7)] is $h(s) = q_\omega(-s)$ in our normalisation; and his mass functions are the left-continuous representatives [4, §1], ours right-continuous — immaterial, since every statement used below concerns continuity points.

2. SETUP

Notation follows [15, 13]: $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, entire of order one, $\xi(s) = \xi(1-s) = \overline{\xi(\bar{s})}$; zeros $\rho = \beta + i\tau$ with $0 \leq \beta \leq 1$, $|\tau| \geq 14.13$; $\Theta = \sup_\rho \beta$, $\vartheta = \Theta - \frac{1}{2}$. Throughout, $\omega > 0$ and $\sigma = \frac{1}{2} + \omega$; the standing hypothesis is

$$\beta < \sigma \quad \text{for every zero } \rho = \beta + i\tau. \quad (7)$$

For $\sigma \geq 1$ this holds unconditionally: every individual zero has $\beta < 1$ ($\beta \leq 1$ trivially, and there are no zeros on $\text{Re}(s) = 1$, by Hadamard and de la Vallée Poussin). For $\sigma < 1$ it is the zero-free half-plane hypothesis $\sigma > \Theta$. We emphasise that at $\sigma = 1$ no uniform gap $\inf_\rho(\sigma - \beta) > 0$ is asserted — it is not known that $\Theta < 1$ — and no proof below requires one: only the pointwise inequality (7) enters.

For strings we use the normalisation of [15, §9]: a string $S[m, L]$ is $0 < L \leq \infty$ together with a right-continuous non-decreasing $m \geq 0$ on $[0, L)$, $m(0-) = 0$; the string equation is $dy' + zy dm = 0$ with solutions $\varphi(\cdot, z)$, $\psi(\cdot, z)$ normalised by $\varphi(0) = \psi'(0) = 1$, $\varphi'(0) = \psi(0) = 0$; the principal Weyl function is $q(z) = \lim_{x \rightarrow L} \psi(x, z)/\varphi(x, z) \in \mathcal{N}_S$, and Krein's theorem [9, 3] says $S[m, L] \mapsto q$ is a bijection onto $\mathcal{N}_S$. The model case used twice below: for $dm = \rho dx$ on $[0, \infty)$, $\rho > 0$ constant,

$$\varphi(x, -y) = \cosh(\sqrt{\rho y} x), \quad \psi(x, -y) = \frac{\sinh(\sqrt{\rho y} x)}{\sqrt{\rho y}}, \quad q(-y) = \frac{1}{\sqrt{\rho y}} \quad (y > 0). \quad (8)$$

Two facts from [15, 13] are used as inputs. First, the Laplace identity (1), valid for $\text{Im } z > \max(0, \frac{1}{2} - \omega)$ and extending by analytic continuation to $\text{Im } z > 0$ under (7), since then $\Psi_\omega(t) = O(t)$ by (2). Second, the Hadamard expansion in the paired form:

Lemma 2.1. *For s off the zeros,*

$$\left(\frac{\xi'}{\xi}\right)(s) = \sum_{\tau > 0} \left[\frac{1}{s - \rho} + \frac{1}{s - \bar{\rho}} \right],$$

the sum over conjugate pairs converging absolutely and locally uniformly, with no additive constant.

Proof. By Hadamard, $(\xi'/\xi)(s) = B + \sum_{\rho} [(s - \rho)^{-1} + \rho^{-1}]$ with $B = -\sum_{\rho} \text{Re}(1/\rho)$, the last sum paired over conjugates ([1, §12]). Grouping conjugate pairs, $\sum_{\text{pairs}} (\rho^{-1} + \bar{\rho}^{-1}) = \sum_{\rho} \text{Re}(1/\rho)$ cancels B exactly. Each paired term is $2(s - \beta)/((s - \beta)^2 + \tau^2) = O(|s|/\tau^2)$, and $\sum \tau^{-2} < \infty$. $\square$

3. THE SPECTRAL MEASURE: PROOF OF THEOREM 1.1

The analytic heart is one residue computation, which identifies, pair by pair, the Hadamard expansion of q_ω with the Stieltjes transform of the corresponding pair of Poisson kernels.

Lemma 3.1. *Let $a > 0$, $\tau \in \mathbb{R}$, $z \in \mathbb{C} \setminus [0, \infty)$, and $w = -i\sqrt{z}$ as in (3). Then*

$$\frac{1}{w} \left[\frac{1}{a + w - i\tau} + \frac{1}{a + w + i\tau} \right] = \frac{1}{\pi} \int_{\mathbb{R}} \left[\frac{a}{a^2 + (u - \tau)^2} + \frac{a}{a^2 + (u + \tau)^2} \right] \frac{du}{u^2 - z}. \quad (9)$$

Proof. Write $v = \sqrt{z}$ ($\text{Im } v > 0$). The integrand decays like u^{-4} ; close the contour in the upper half u -plane. Because $a > 0$, the Lorentzian poles $u = \pm\tau + ia$ lie in the *upper* half-plane and contribute $((\pm\tau + ia)^2 - z)^{-1}$; the pole $u = v$ contributes $(i/v) a (a^2 + (v \mp \tau)^2)^{-1}$. Summing the four residues, both sides of (9) are rational in (a, τ, v) ; clearing denominators reduces the identity to

$$(a^2 + v^2 + \tau^2)^2 - 4v^2\tau^2 = (\tau^2 - a^2 - v^2)^2 + 4a^2\tau^2 \quad (\text{both} = (a^2 + v^2)^2 + 2\tau^2(a^2 - v^2) + \tau^4)$$

with matching numerators $-2(a^2 + v^2 - \tau^2)$, an elementary verification. (It was also checked numerically at four parameter triples, including the near-cut point $z = 100 + i$, to 10^{-33} – 10^{-44} .) $\square$

Proof of Theorem 1.1. Analyticity and reality. For $z \in \mathbb{C} \setminus [0, \infty)$ we have $\text{Re } w > 0$, so $\text{Re}(\sigma + w) > \sigma$; by (7) the open half-plane $\text{Re}(s) > \sigma$ contains no zeros, and q_ω is analytic on the cut plane. On $(-\infty, 0)$, $w = \sqrt{|z|} > 0$ and

$$q_\omega(-y) = \frac{(\xi'/\xi)(\sigma + \sqrt{y})}{\sqrt{y}} > 0 \quad (y > 0), \quad (10)$$

positivity because each paired Hadamard term $2(s - \beta)/((s - \beta)^2 + \tau^2)$ is positive for real $s \geq \sigma$, by (7).

The Stieltjes-transform identity. Apply Lemma 3.1 to each conjugate pair, with $a = \sigma - \beta > 0$ — which is exactly the standing hypothesis (7), and requires no uniform lower bound. Summing over pairs and using Lemma 2.1 on the left,

$$q_\omega(z) = \int_{\mathbb{R}} \frac{\rho_\omega(u)}{u^2 - z} du = \int_0^\infty \frac{d\sigma_\omega(\lambda)}{\lambda - z}, \quad (11)$$

the second equality by $\lambda = u^2$ and evenness of ρ_ω . The interchange of sum and integral is dominated convergence: $|u^2 - z| \geq c_z(1 + u^2)$ for fixed z off the cut, and the convolution of two Poisson kernels being a Poisson kernel,

$$\int_{\mathbb{R}} \frac{1}{\pi} \frac{a}{a^2 + (u - \tau)^2} \cdot \frac{du}{1 + u^2} = \frac{1 + a}{(1 + a)^2 + \tau^2} = O\left(\frac{\sigma}{\tau^2}\right),$$

which is summable over the zeros.

Stieltjes class and the representation. By (11) with $\rho_\omega \geq 0$: $\text{Im } q_\omega(z) = \text{Im } z \int \rho_\omega(u) |u^2 - z|^{-2} du > 0$ on $\mathbb{C}^+$, and $q_\omega > 0$ on $(-\infty, 0)$, so $q_\omega \in \mathcal{N}_S$. The measure satisfies $\int (1 + \lambda)^{-1} d\sigma_\omega < \infty$: near 0 the density is $\sim \rho_\omega(0)/\sqrt{\lambda}$, integrable; at ∞ it is $O(\log \lambda \cdot \lambda^{-1/2}) \cdot \lambda^{-1}$ against $d\lambda$, again integrable, since $\rho_\omega(u) = O(\log u)$ uniformly for $\sigma \geq 1$: by Stirling $\frac{1}{2}\psi(\frac{\sigma+iu}{2}) = \frac{1}{2}\log \frac{u}{2} + O(1)$, and $(\zeta'/\zeta)(\sigma + iu) = O(\log u)$ classically [16, (3.11.7)]. (For fixed $\sigma > 1$, absolute convergence of the Dirichlet series gives the sharper pointwise $\rho_\omega(u) = \frac{1}{2\pi} \log \frac{u}{2\pi} + O_\sigma(1)$; uniformly down to $\sigma = 1$ only the $O(\log u)$ bound is claimed, and only it is needed. In the mean, ρ_ω follows the Riemann–von Mangoldt density — see Figure 1.) The additive constant is $b = \lim_{y \rightarrow \infty} q_\omega(-y) = 0$ by (12), and there is no linear term.

Absolute continuity. ξ'/ξ is analytic at every point of the closed half-plane $\text{Re}(s) \geq \sigma$: the zeros are discrete, none lies there by (7), and the pole of ζ at $s = 1$ is cancelled in ξ ($\xi(1) = \frac{1}{2} \neq 0$). So q_ω extends continuously from $\mathbb{C}^+$ to $(0, \infty)$ with

$$\frac{1}{\pi} \text{Im } q_\omega(\lambda + i0) = \frac{1}{\sqrt{\lambda}} \cdot \frac{1}{\pi} \text{Re} \left(\frac{\xi'}{\xi} \right) (\sigma + i\sqrt{\lambda}) = \frac{\rho_\omega(\sqrt{\lambda})}{\sqrt{\lambda}},$$

using $w \rightarrow -i\sqrt{\lambda}$ and conjugate symmetry. Stieltjes–Perron inversion on compact subsets of $(0, \infty)$ identifies $d\sigma_\omega$ with this continuous density; there is no singular part on $(0, \infty)$, and no atom at 0 since $q_\omega(z) = O(|z|^{-1/2})$ near 0 gives $\varepsilon |q_\omega(i\varepsilon)| \rightarrow 0$.

The string. Kreĭn's theorem [9, 3] attaches the unique $S[m_\omega, L_\omega]$. By (10), $q_\omega(-y) \uparrow \infty$ as $y \downarrow 0$ (the transform $\int d\sigma_\omega/(\lambda + y)$ is decreasing in y and diverges, the measure having a $\lambda^{-1/2}$ density at 0); since the length is $L_\omega = \lim_{s \downarrow 0} h(s)$ in the Kreĭn–Kasahara normalisation [4, (9)], with $h(s) = q_\omega(-s)$, this gives $L_\omega = \infty$ directly; and Theorem 1.3 gives $m_\omega(\infty) = \infty$. $\square$

Remark 3.2 (where positivity enters). The proof localises the role of the zero-free half-plane exactly. If a zero has $\beta > \sigma$, its Lorentzian pole $u = \tau + i(\sigma - \beta)$ crosses into the *lower* half u -plane and Lemma 3.1 acquires a residue correction: q_ω then equals $\int \rho_\omega(u)(u^2 - z)^{-1} du$ plus explicit non-Herglotz pole terms. The boundary formula for ρ_ω in (4) continues to hold, but q_ω is no longer the Stieltjes transform of its own boundary density. This is the mechanism behind Theorem 1.5; whether the density ρ_ω itself becomes negative somewhere below the critical shift is a separate question, which we neither need nor claim (a violating zero at height τ contributes $-1/(\beta - \sigma)$ at $u = \tau$ against a positive background of order $\log \tau$).

4. THE SCREW REPRESENTATION: PROOF OF THEOREM 1.2

Proof of Theorem 1.2. Both sides of (5) are continuous in t and polynomially bounded: for the right side, $(1 - \cos ut)u^{-2} \leq \min(t^2/2, 2u^{-2})$ and $\int_{\mathbb{R}} \min(t^2, u^{-2}) \rho_{\omega}(u) du < \infty$ since $\int_{|u|>1} \rho_{\omega}(u)u^{-2}du < \infty$; for the left side this is (2). So both Laplace transforms converge on $\text{Im } z > 0$, and by uniqueness of Laplace transforms of continuous polynomially bounded functions it suffices to match them. The left transform is (1). For the right, Fubini (justified by the same min-bound against $e^{-t\text{Im } z}$) and the elementary transform

$$\int_0^{\infty} (1 - \cos ut) e^{izt} dt = \frac{-iu^2}{z(z^2 - u^2)} \quad (\text{Im } z > 0)$$

give

$$\int_0^{\infty} \left[\int_{\mathbb{R}} \frac{1 - \cos ut}{u^2} \rho_{\omega}(u) du \right] e^{izt} dt = \frac{i}{z} \int_{\mathbb{R}} \frac{\rho_{\omega}(u)}{u^2 - z^2} du = \frac{i}{z} q_{\omega}(z^2),$$

by (11) at $Z = z^2 \in \mathbb{C} \setminus [0, \infty)$. For $\text{Im } z > 0$ the principal upper root of z^2 is z itself, so by (3) $q_{\omega}(z^2) = (\xi'/\xi)(\sigma - iz)/(-iz)$, and the transform equals $-z^{-2}(\xi'/\xi)(\sigma - iz)$ — identical to (1).

The margin identity (6) is then immediate: $\rho_{\omega}(0) = \pi^{-1}(\xi'/\xi)(\sigma)$, since ξ'/ξ is real at real arguments. Its content is Theorem 1.1: the function whose value at 0 is the margin slope is the spectral density of the string. $\square$

Remark 4.1. The normalisation in (5) is fixed twice over: by the transform computation above, and independently by a numerical comparison against Suzuki's arithmetic (prime-side) formula for $\Psi_{1/2}$ [15, §11], which agrees to relative accuracy $1.3 \cdot 10^{-6}$ at $t = 2$ (Section 7). Note also that “zero-frequency density” in (6) refers to the density ρ_{ω} on the frequency line u ; in the spectral variable $\lambda = u^2$ the density diverges as $\rho_{\omega}(0)/\sqrt{\lambda}$.

5. BULK AND ULTRAVIOLET LAWS: PROOFS OF THEOREMS 1.3 AND 1.4

Both proofs consist of an exact asymptotic computation on the Weyl function — available in closed form by (10) — followed by Kasahara's regular-variation theorem for Kreĭn's correspondence, [4, Thm. 2], applied at index $\alpha = \frac{1}{2}$. We use his part (iii): for $0 < \alpha \leq 1$ and L continuous, slowly varying at ∞ [0], with $t^{\alpha}L(t)$ increasing for large [small] t and inverse function f_{α} , the asymptotic $h(s) \sim D_{\alpha}s^{-\alpha}L(1/s)$ as $s \rightarrow 0^+$ [∞] is equivalent to $m(x) \sim x^{1/\alpha-1}K_{\alpha}(x)$ as $x \rightarrow \infty$ [0^+], where $K_{\alpha}(x) = x^{-1/\alpha}f_{\alpha}(x)$ is slowly varying and D_{α} is the constant of his Example 1 — at $\alpha = \frac{1}{2}$, $D_{1/2} = \{\frac{1}{4}\}^{-1/2}\Gamma(\frac{3}{2})/\Gamma(\frac{1}{2}) = 1$. Here $h(s) = q_{\omega}(-s)$ is the characteristic function of m_{ω} [4, (5),(7)].

Proof of Theorem 1.3. From (10),

$$q_{\omega}(-y) \sqrt{y} = \left(\frac{\xi'}{\xi}\right)(\sigma + \sqrt{y}) \longrightarrow \left(\frac{\xi'}{\xi}\right)(\sigma) > 0 \quad (y \downarrow 0),$$

with error $O(\sqrt{y})$; positivity of the limit is Lemma 2.1 under (7) (quantitatively, for $0 < \omega < \frac{1}{2}$, $(\xi'/\xi)(\sigma) \geq (2 + \gamma_0 - \log 4\pi)\omega$ [13, Cor. 3.1]). So the characteristic function satisfies $h(s) = q_{\omega}(-s) \sim D_{1/2}s^{-1/2}L(1/s)$ as $s \downarrow 0$, with $D_{1/2} = 1$ and the constant slowly varying function $L \equiv c := (\xi'/\xi)(\sigma)$; $t^{1/2}L(t) = c\sqrt{t}$ is increasing, so the hypotheses of [4, Thm. 2] hold at its unbracketed ($s \rightarrow 0^+$, $x \rightarrow \infty$) endpoint. Part (iii), (A) $\Leftrightarrow$ (C), gives

$$m_{\omega}(x) \sim x K_{1/2}(x) \quad (x \rightarrow \infty), \quad K_{1/2}(x) = x^{-2}f_{1/2}(x) = x^{-2}(x/c)^2 = c^{-2},$$

$f_{1/2}$ being the inverse of $t \mapsto c\sqrt{t}$. Hence $m_\omega(x)/x \rightarrow c^{-2}$. (Sanity check: for the homogeneous string (8) both asymptotics are identities, consistent with $D_{1/2} = 1$.) The numerical value at $\omega = \frac{1}{2}$ uses the classical $(\xi'/\xi)(1) = -(\xi'/\xi)(0) = \frac{\gamma_0}{2} + 1 - \frac{1}{2} \log 4\pi$. $\square$

Proof of Theorem 1.4. At the other end, for $y \rightarrow \infty$,

$$\left(\frac{\xi'}{\xi}\right)(\sigma + \sqrt{y}) = \frac{1}{2}\psi\left(\frac{\sigma + \sqrt{y}}{2}\right) - \frac{1}{2} \log \pi + O(y^{-1/2}) = \frac{1}{4} \log y (1 + O(1/\log y)), \quad (12)$$

since $(\xi'/\xi)(\sigma + \sqrt{y}) = O(2^{-\sqrt{y}})$; note the leading term is ω -independent and unconditional — this is the analytic face of the fact that the Riemann–von Mangoldt density does not see the shift. Hence the characteristic function satisfies $h(s) = q_\omega(-s) \sim D_{1/2} s^{-1/2} L(1/s)$ as $s \rightarrow \infty$, with $D_{1/2} = 1$ and $L(t) = \frac{1}{4} \log(1/t)$: L is continuous and slowly varying at 0, and $t^{1/2} L(t) = \frac{1}{4} \sqrt{t} \log(1/t)$ is increasing for small t , so the hypotheses of [4, Thm. 2] hold at its bracketed $(s \rightarrow \infty, x \rightarrow 0^+)$ endpoint. Part (iii), (A) $\Leftrightarrow$ (C), gives $m_\omega(x) \sim x K_{1/2}(x)$ as $x \rightarrow 0^+$, with $K_{1/2}(x) = x^{-2} f_{1/2}(x)$ and $f_{1/2}$ the inverse, near 0, of $t \mapsto \frac{1}{4} \sqrt{t} \log(1/t)$. Inverting,

$$\sqrt{f_{1/2}(x)} = \frac{4x}{\log(1/f_{1/2}(x))} \sim \frac{2x}{\log(1/x)}, \quad \text{so} \quad f_{1/2}(x) \sim \frac{4x^2}{(\log x)^2}, \quad K_{1/2}(x) \sim \frac{4}{(\log x)^2},$$

and $m_\omega(x) \sim 4x/(\log x)^2$. The constant agrees with Kotani's RH-conditional case [7] because only the leading Riemann–von Mangoldt growth of the spectral measure enters, and that is the same for σ_ω as for Kotani's pure-point measure. $\square$

Remark 5.1 (the ultraviolet law is invisible to feasible numerics). On the analytic dual curve at $\omega = \frac{1}{2}$, the ratio of $M(y)$ to $4x/(\log x)^2$ (at $x = q(-y)$) is

$$374, \quad 4.26, \quad 1.388, \quad 1.093, \quad 1.008, \quad 0.974 \quad \text{at } y = 1, 10^3, 10^6, 10^9, 10^{12}, 10^{15},$$

tending to 1 non-monotonically, with slowly varying correction $[(\frac{1}{2} \log y - \log(\xi'/\xi))/(\frac{1}{2} \log y - \log 2\pi)]^2$. Probing $y = 10^{12}$ requires resolving the spectral measure to $u \sim 10^6$; no string reconstruction of feasible size sees this regime, which is why Theorem 1.4 is an analytic statement and the numerics of Section 7 validate only the crossover and bulk regions.

6. THE ENDPOINT: PROOFS OF THEOREMS 1.5 AND 1.6

Proof of Theorem 1.5. $q_\omega(z) = (\xi'/\xi)(\sigma + w)/w$ has a pole wherever $\sigma + w = \rho$, i.e. $w = (\beta - \sigma) + i\tau$, which lies in the admissible half-plane $\operatorname{Re} w > 0$ exactly when $\beta > \sigma$. Then $z = -w^2 = \tau^2 - (\beta - \sigma)^2 - 2i\tau(\beta - \sigma)$, which is non-real for $\tau \neq 0$ (and there are no real zeros of ξ); the conjugate zero $\bar{\rho}$ gives the conjugate pole. A Herglotz function is analytic off $\mathbb{R}$, so a non-real pole in the cut plane already excludes q_ω from the Herglotz class, a fortiori from $\mathcal{N}_S$; by Kreĭn's bijection no string has q_ω as Weyl function. The displayed equivalence combines this with the conditional case of Theorem 1.1: RH gives (7) for every $\omega > 0$; conversely, a zero with $\beta > \frac{1}{2}$ destroys $\mathcal{N}_S$ for every $\omega < \beta - \frac{1}{2}$. At $\omega = \omega_c$ itself the family survives; see Remark 6.1. $\square$

Remark 6.1 (the critical member). Theorem 1.5 says the family cannot be continued *below* $\omega_c = \Theta - \frac{1}{2}$; at $\omega = \omega_c$ it still exists. If the supremum Θ is not attained, every zero satisfies $\beta < \sigma_c = \frac{1}{2} + \omega_c$, hypothesis (7) holds, and Theorem 1.1 applies verbatim. If it is attained, the extremal zeros sit on the boundary line and contribute *real* poles on the cut — which a

Stieltjes function is permitted to have: the contribution of a boundary pair $\{\sigma_c + i\tau, \sigma_c - i\tau\}$ of multiplicity m to the paired Hadamard expansion of q_{ω_c} is exactly

$$\frac{m}{w} \left[\frac{1}{w - i\tau} + \frac{1}{w + i\tau} \right] = \frac{2m}{w^2 + \tau^2} = \frac{2m}{\tau^2 - z},$$

a point mass $2m$ at $\lambda = \tau^2$ — the degenerate $a \rightarrow 0^+$ case of Lemma 3.1, with the same mass normalisation as Kotani's string. The critical member is thus a string whose spectral measure carries point masses $2m_\rho$ at the squared ordinates of the extremal zeros, embedded in the absolutely continuous background of the remaining zeros. (Consistently, on the time side the shifted criterion also survives at the critical shift [13].) Moreover, for $\omega_c > 0$ the bulk and ultraviolet laws persist at criticality: the closed form (10) is unchanged at $\omega = \omega_c$, and $(\xi'/\xi)(\sigma_c) > 0$ still — an extremal pair contributes 0 to the paired Hadamard sum at real σ_c , while its mirror pair on $\text{Re}(s) = 1 - \sigma_c$ contributes $4\omega_c/(4\omega_c^2 + \tau^2) > 0$ — so the proofs of Theorems 1.3 and 1.4 apply verbatim to the critical string: $m_{\omega_c}(x) \sim [(\xi'/\xi)(\sigma_c)]^{-2}x$ at infinity and $\sim 4x/(\log x)^2$ at 0. The distinction at criticality is thus purely one of spectral type — absolutely continuous, versus absolutely continuous plus embedded atoms. (The case $\omega_c = 0$, where the mirror symmetry makes the slope vanish, is genuinely different: see Remark 6.3.) The finer structure of the critical string is left as an open problem (Section 9).

Proof of Theorem 1.6. (i) For $K \subset \mathbb{C} \setminus ([0, \infty) \cup P)$ compact, the w -image of K has $\text{Re } w \geq \delta(K) > 0$, so for $0 \leq \omega \leq \omega_0$ the arguments $\sigma + w$ range in a fixed compact set on which ξ'/ξ is analytic (avoiding, by the definition of P , the finitely many relevant zeros). Then, by the fundamental theorem of calculus along the real segment from $\frac{1}{2} + w$ to $\frac{1}{2} + \omega + w$,

$$q_\omega(z) - q_\xi(z) = \frac{1}{w} \int_0^\omega \left(\frac{\xi'}{\xi} \right)' \left(\frac{1}{2} + t + w \right) dt, \quad \text{whence} \quad |q_\omega(z) - q_\xi(z)| \leq \frac{\omega}{\inf_{z \in K} |w|} \sup_{\substack{z \in K \\ 0 \leq t \leq \omega_0}} \left| \left(\frac{\xi'}{\xi} \right)' \left(\frac{1}{2} + t + w \right) \right|$$

the locally uniform $O(\omega)$ rate. (The integral form is used because the one-variable mean value theorem is not available for complex-valued functions.) Under RH, every pole of q_ξ sits at $z = \tau^2 \in [0, \infty)$ (as $w = i\tau$ has $\text{Re } w = 0$), so $P \subset [0, \infty)$.

(ii) Under RH, (4) reads $\rho_\omega(u) = \frac{1}{\pi} \sum_\tau \omega/(\omega^2 + (u - \tau)^2)$, the sum over all ordinates (both signs): the density is the Poisson broadening of the zero set, and the claim is the approximate-identity limit, summed over the zeros. Let $f \in C_c(\mathbb{R})$ with $\text{supp } f \subset [-R, R]$. Each term satisfies $\int_{\mathbb{R}} P_\omega(u - \tau) f(u) du \rightarrow f(\tau)$ as $\omega \downarrow 0$, where P_ω is the unit-mass Poisson kernel. The terms are dominated, uniformly in $0 < \omega \leq 1$, by $\min(\|f\|_\infty, \pi^{-1} \|f\|_{L^1} d(\tau)^{-2})$ with $d(\tau) = \text{dist}(\tau, [-R, R])$, which is summable over the zeros: only finitely many ordinates have $d(\tau) \leq 1$, and $\sum \tau^{-2} < \infty$. Dominated convergence for the series gives $\int f d\mu_\omega \rightarrow \sum_\tau f(\tau)$, i.e. $\mu_\omega \rightarrow \sum_\tau \delta_\tau$ vaguely on $\mathbb{R}$; pushing forward under $u \mapsto u^2$ (for $g \in C_c([0, \infty))$, $g(u^2) \in C_c(\mathbb{R})$) yields $\sigma_\omega \Rightarrow \sum_\tau \delta_{\tau^2} = 2 \sum_{\tau > 0} \delta_{\tau^2}$, which is Kotani's measure with its normalisation [15, §9]. (One may instead argue abstractly from (i) via convergence theorems for Herglotz functions, but these require care here, both measures having infinite total mass; the direct proof makes the sharpening of the Lorentzians literally the mechanism.)

(iii) Kasahara's continuity theorem [4, Thm. 1] states that, for strings m_n with characteristic functions $h_n(s) = q_n(-s)$, convergence $m_n(x) \rightarrow m_0(x)$ at every continuity point of m_0 is equivalent to $h_n(s) \rightarrow h_0(s)$ for every $s > 0$. By (i) and RH, $q_\omega(-s) \rightarrow q_\xi(-s)$ for every $s > 0$ — the negative axis lies in $\mathbb{C} \setminus [0, \infty)$, away from the poles of q_ξ — along every sequence $\omega_n \downarrow 0$; hence $m_\omega(x) \rightarrow m_\xi(x)$ at every continuity point of m_ξ . (Kasahara works with left-continuous mass functions [4, §1], ours are right-continuous; the representatives agree off the countable jump set, and the statement concerns continuity points only.) $\square$

Remark 6.2. Vague (local weak) convergence is the correct topology in (ii): both σ_ω and the limit have infinite total mass, and mass escapes to $\lambda = \infty$ under any global weak formulation. Quantitatively, the convergence in (i) is measured (Section 7) as exactly linear in ω over $\omega \in [0.01, 0.5]$ at fixed test points, and in (ii) a unit Gaussian test function centred at the first ordinate is captured to $1 - O(\omega)$.

Remark 6.3 (the endpoint length collapse). The length of the string detects the endpoint as sharply as the spectral type. For every $\omega > 0$, $L_\omega = \lim_{s \downarrow 0} q_\omega(-s) = \infty$ (Theorem 1.1(iv), via [4, (9)]), because $(\xi'/\xi)(\sigma) > 0$. Under RH, at the endpoint itself the same formula gives, by the oddness of ξ'/ξ about $\frac{1}{2}$,

$$q_\xi(-s) = \frac{(\xi'/\xi)(\frac{1}{2} + \sqrt{s})}{\sqrt{s}} \longrightarrow \left(\frac{\xi'}{\xi}\right)'(\frac{1}{2}) = B(0) \quad (s \downarrow 0),$$

so Kotani's string has *finite* length

$$L_\xi = B(0) = \sum_{\gamma} \gamma^{-2} = 0.0462099862 \dots$$

— the margin-scale constant of [13, Rem. 2.3]; equivalently $L_\xi = q_\xi(0^-) = \sum_{\tau > 0} 2/\tau^2$ from Suzuki's expansion [15, §9] (numerically, $q_\xi(-10^{-10}) = 0.0462099862$). Under RH the $\omega \downarrow 0$ limit is therefore doubly singular: the spectrum degenerates from absolutely continuous to pure point, and the geometry collapses from infinite-length strings of homogeneous bulk density $[(\xi'/\xi)(\frac{1}{2} + \omega)]^{-2} \sim (B(0)\omega)^{-2}$ to a finite string of length $B(0)$. The infinitely dense bulk does not persist in the limit: in the inextensible normalisation $m_\xi \equiv \infty$ on $(B(0), \infty)$, so Theorem 1.6(iii) forces $m_\omega(x) \rightarrow \infty$ for every $x > B(0)$ — the bulk mass is expelled beyond the finite length as $\omega \downarrow 0$.

7. NUMERICAL COMPUTATION OF THE STRINGS

All computations are in `mpmath`; scripts are available from the author. Six independent checks of Theorem 1.1 were run: the closed form of ρ_ω against the Lorentzian zero-sum (1000 zeros plus a Riemann–von Mangoldt tail; relative difference 10^{-5} – 10^{-7}); against the prime side of ξ'/ξ at $\sigma = 2$ ($2 \cdot 10^{-7}$); positivity on $u \in [0, 200]$ at $\omega = \frac{1}{2}$ (minimum $= \rho_{1/2}(0) = 0.0073516$ at $u = 0$); the integrated density against the Riemann–von Mangoldt count ($\int_0^{100} \rho_{1/2} = 29.111$ vs. $N(100) = 29.002$ — agreement at the smearing level; the integrated spectral measure has the same leading growth $\frac{u}{2\pi} \log \frac{u}{2\pi e}$ as the zero-counting function); identity (5) against Suzuki's arithmetic $\Psi_{1/2}$ ($1.3 \cdot 10^{-6}$ at $t = 2$); and (11) at complex test points ($4 \cdot 10^{-7}$). The residue identity (9) was verified separately to 10^{-33} – 10^{-44} .

7.1. The discrete inverse problem. The measure $d\sigma_\omega$ is discretised by Gauss–Legendre panels (≤ 152 point masses up to a spectral cutoff), and the unique finite Stieltjes string with the resulting rational Weyl function is recovered by Herglotz peeling — repeatedly stripping the leading mass and length read off the large- y behaviour and passing to the inverted remainder, with bracketed root-finding (one root per gap, guaranteed by interlacing) at 90–120 decimal digits. Validation: synthetic round trips are exact to 10^{-121} , and every production string reproduces its target q at complex test points to $\leq 10^{-117}$, with all masses and lengths positive. We emphasise the scope of these figures: they certify the solution of the *discrete* inverse problem — the inversion of the finite discretised measure — to essentially exact precision. The distance from the finite string to the continuum ζ -string is a separate,

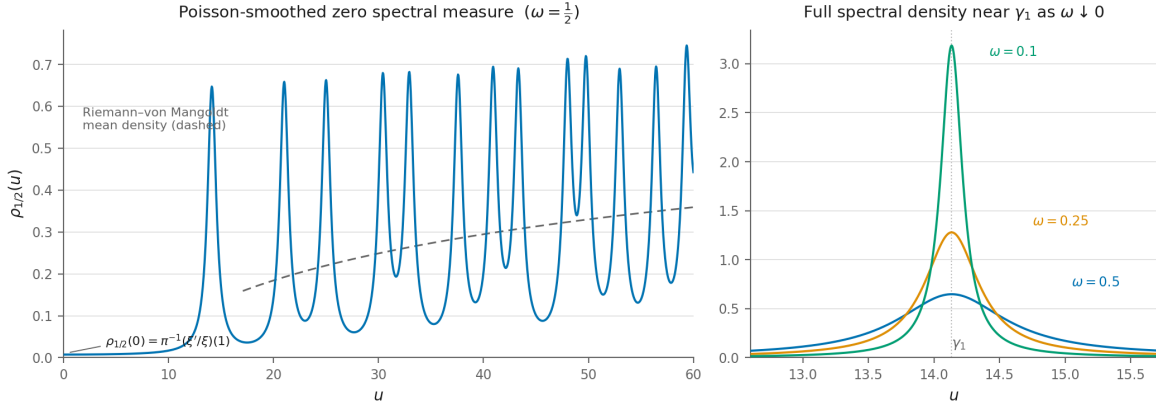


FIGURE 1. The Poisson-smoothed zero spectral measure. Left: the density $\rho_{1/2}(u)$ of (4) on $0 \leq u \leq 60$ (solid), against the Riemann–von Mangoldt mean density $\frac{1}{2\pi} \log \frac{u}{2\pi}$ (dashed). Under RH each zero contributes a Lorentzian of width ω ; for the plotted low zeros this is unconditionally valid, their critical-line location being verified [14]. The zero-frequency value is the margin slope of [13] divided by π (Theorem 1.2). Right: the full closed-form density $\rho_\omega(u)$ near the first ordinate for $\omega = 0.5, 0.25, 0.1$, sharpening toward the point mass of Kotani’s string as $\omega \downarrow 0$ (Theorem 1.6); the curves are visually indistinguishable from the first zero’s own Poisson kernel because the background of all other zeros contributes only 0.011, 0.005, 0.002 at the peak (against peak values 0.647, 1.279, 3.185; the bare kernel peaks at $1/\pi\omega$). For $\omega < \frac{1}{2}$ the local picture at this zero is unconditional — its critical-line location is verified — but the interpretation of ρ_ω as the spectral density of a positive string at those shifts is conditional (Theorem 1.5).

quadrature-dominated question, which the flat-control comparison below is designed to cancel to leading order.

7.2. Infrared bulk. Raw secant fits of x/m_ω^{-1} deep in the reconstructed bulk give

$$\frac{\text{fitted } \rho_\infty}{\text{predicted } (\xi'/\xi)(\sigma)^{-2}} = 1.30760, 1.30774, 1.30776 \quad \text{at } \omega = 0.5, 0.75, 1.0 :$$

the ω -scaling of Theorem 1.3 holds to $\sim 10^{-4}$, and the common factor 1.308 is discretisation bias, measured on an exactly solvable homogeneous control string discretised on the same grid as 1.30826. Control-corrected absolute agreement: 0.9995, 0.9996, 0.9996. This is validation of Theorem 1.3, not its proof. At $\omega = \frac{1}{2}$ the predicted unconditional bulk density is $\rho_\infty = 1874.7243\dots$

7.3. Profile. The bias-cancelled ratio $m_\zeta(x)/m_{\text{flat}}(x)$ rises from ≈ 0 in a light, Kotani-type ultraviolet region, through an arithmetic crossover at $x \approx 0.05$ –1 — dual to the lowest zeros, $x \sim q(-\tau_1^2) \approx 0.05$ — to the homogeneous bulk (equal to 1 within 10^{-3} for $x \gtrsim 3$). The crossover location is the one feature of the profile with a clean analytic anchor; the ultraviolet law itself is not numerically reachable (Remark 5.1).

8. THE PHYSICS DICTIONARY

The mathematics above supports the following dictionary, stated last deliberately; entries are labelled by their standing.

- (*thm*) Zeros $\leftrightarrow$ resonant spectral locations: each zero is a Lorentzian resonance of the density of states, of width its distance to the boundary line — width exactly ω under RH (Theorem 1.1).
- (*thm*) $\omega \leftrightarrow$ broadening/damping scale; the margin slope of [13] $\leftrightarrow \pi \times$ zero-frequency density of states (Theorem 1.2).
- (*thm*) Bulk string density $\leftrightarrow$ inverse square of the margin slope (Theorem 1.3): the central quantity of the shifted screw criterion is, on the string, the infrared bulk density — equivalently, the margin slope is the bulk “sound speed” $\rho_\infty^{-1/2}$.
- (*thm*) $\omega \downarrow 0$ under RH $\leftrightarrow$ an absolutely-continuous-to-pure-point spectral limit onto Kotani’s string (Theorem 1.6); $\rho_\infty \sim (B(0)\omega)^{-2} \rightarrow \infty$: the bulk becomes infinitely heavy — and simultaneously the geometry collapses, $L_\omega = \infty$ for every $\omega > 0$ while $L_\xi = B(0) < \infty$ (Remark 6.3): the limit is not an infinitely dense infinite bulk, but a finite string of length $B(0)$ with the bulk mass expelled beyond it.
- (*thm*) Failure of the zero-free half-plane $\leftrightarrow$ failure of the Herglotz/passive-realisation property, through explicit non-real poles (Theorem 1.5); on the time side, Ψ_ω then oscillates two-sidedly at exponential rate (the critical-shift dichotomy, [13, Thm. 3.4]).
- (*gloss*) Words such as “localisation transition” or “phase transition” are interpretive glosses on the above: no statement about eigenfunctions or dynamics is claimed, and the order-parameter language is made precise, on the time side only, in [13].

9. OPEN PROBLEMS

- (1) *Effective endpoint convergence.* Theorem 1.6(iii) is qualitative. An effective rate for $m_\omega \rightarrow m_\xi$ (under RH), even at fixed x , would quantify how the arithmetic crossover region of the string records the low zeros.
- (2) *The critical member.* When Θ is attained, Remark 6.1 exhibits q_{ω_c} as the Weyl function of a string whose spectral measure has point masses embedded in an absolutely continuous background, and shows that the bulk and ultraviolet laws persist at ω_c . What remains open is finer: the local structure of the critical string at the embedded atoms, and whether anything on the string side distinguishes the attained from the non-attained case.
- (3) *Direct high-resolution inversion.* The $O(n^3)$ peeling limits the discretisation to ~ 150 masses. Resolving the bulk directly (rather than by control cancellation) needs node spacings $\delta u \approx \pi/(X\sqrt{\rho_\infty})$, i.e. 10^3 – 10^4 masses; a Gelfand–Levitan or Padé/Kasahara route would do better and could turn the absolute bulk constant into a sharp numerical test.
- (4) *The family aspect.* The final remarks of [13] indicate that analogous shifted formulas hold for self-dual Dirichlet L -functions, and that off-central real zeros behave differently, their contribution carrying no oscillatory phase. Constructing the corresponding ω -strings, understanding their conductor dependence (the bulk density falls as the margin grows), and analysing the effect of exceptional real zeros are natural extensions.
- (5) *Other realisations.* The relation of the ω -string family at $\omega \downarrow 0$ to the de Branges-space realisations studied by Lagarias [12], and to Kotani’s construction [7], in a common inverse-spectral topology, deserves a careful treatment; Theorem 1.6 is the first step.

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